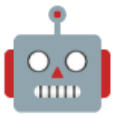

LLMs and How Do They Work

Large Language Models (LLMs): How Do They Work?

1. What *Exactly* Is an LLM?

Large Language Models are **Transformer-based neural networks** trained on vast text corpora to predict the next token in a sequence. Because language exhibits statistical regularities, the model can leverage those patterns to produce fluent, context-aware text.

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2. Core Building Blocks

2.1 Tokenization

- Text is broken into *tokens* (words, sub-words, or bytes).
- Each token is mapped to an **embedding vector**—a dense numerical representation of its contextual meaning.

2.2 The Transformer Architecture

- **Self-Attention Layers** learn relationships between all tokens in a sequence, regardless of distance.
- **Feed-Forward Networks** transform attended information into higher-level features.
- **Residual Connections & Layer Norm** stabilize training and allow very deep stacks of layers (often 24, 48, 96+).

2.3 Parameters & Scale

- Models range from millions to **hundreds of billions of parameters** (learned weights).

- More parameters → greater capacity to capture nuanced language patterns, but with higher compute cost.

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3. Training: Next-Token Prediction at Scale

1. **Objective:** Minimize cross-entropy loss between the model's predicted probability distribution and the *true* next token.
2. **Self-Supervised Data:** Massive, diverse text (books, articles, code, web pages) provides free "labels" because every next token is known.
3. **Optimization:** Back-propagation with stochastic gradient descent variants (Adam, LAMB).
4. **Regularization:** Techniques like dropout, weight decay, and mixed-precision keep training stable.

> Key Point: The model never learns "facts" explicitly; it learns statistical correlations that approximate factual knowledge.

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4. Inference: From Probabilities to Sentences

1. **Prompt Ingestion** → Tokens are embedded and fed through the trained Transformer.
2. **Probability Distribution** → For the next token, the model outputs a softmax distribution over the vocabulary.
3. **Decoding / Sampling** →
   - **Temperature** rescales probabilities (controls randomness).
   - **Top-K** or **Top-P** restrict the candidate pool.
   - A token is **sampled**, appended to the prompt, and the loop repeats.
4. **Iterative Growth** → The sequence grows **token-by-token** until a stop condition (max length, end-of-text token, etc.) is reached.

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5. Why LLMs *Feel* Intelligent

Capability	Underlying Mechanism

Syntax mastery	Self-attention captures long-range grammatical dependencies.
World knowledge	Statistical co-occurrence across billions of sentences.
Few-shot generalization	In-context learning lets the model infer new tasks from examples in the prompt.
Reasoning & chain-of-thought	Sequential prediction can mimic stepwise deduction when guided by well-structured prompts.

6. Limitations & Failure Modes

- **Hallucinations:** Confidently generated but false statements when probability mass favors plausible-sounding text.
- **Context window limits:** Older tokens may be “forgotten” once the window (e.g., 8k, 32k tokens) is exceeded.
- **Bias & toxicity:** Reflections of biased training data can surface in outputs.
- **Arithmetic & logic errors:** Token-level prediction can stumble on tasks requiring strict symbolic precision.
- **Privacy leakage:** Rare or sensitive data from training corpora may inadvertently appear in generations.

7. Practical Prompt-Engineering Tips

1. **System > User > Assistant hierarchy:** Provide a clear *system* role to steer model behavior globally.
2. **Few-shot examples:** Show the model *how* to respond before asking it to generalize.
3. **Chain-of-thought prompts:** Encourage step-by-step reasoning (“Let’s think step by step”).
4. **Explicit constraints:** State output format, length limits, or disallowed content.
5. **Iterative sampling:** Generate multiple completions with varied decoding settings; curate the best.
6. **Post-processing:** Use external validators (e.g., regex, knowledge bases) to fact-check or enforce structure.

8. Key Takeaways

- **LLMs are probabilistic sequence predictors**, not databases or reasoning engines—prompting shapes their behavior.
- **Decoding parameters** (temperature, top-K, top-P) convert probability distributions into human-readable text.
- **Scale + data diversity** grant impressive fluency, but **robust prompting and validation** are required for reliability.

> Bottom line: Understanding token-level prediction, Transformer internals, and decoding strategies empowers you to craft prompts that harness LLM strengths while mitigating their weaknesses.